Continuous Monitoring System

Quality Assurance Program

Cemex Construction Materials South, LLC
Lyons Plant
Main Kiln, Clinker Cooler, Raw Material Dryer Stacks
Lyons, CO

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Overview

This Continuous Monitoring System (CMS) Quality Assurance Program (QAP) is the written operating manual, which incorporates quality assurance/quality control (QA/QC) data quality objectives (DQO), for the CMS associated with the Cemex Lyons Cement Plant. The CMS QAP fulfills the requirements for 40 CFR 63, Section 8(d) and 40 CFR 60, Appendix F. The Environmental Manager is responsible for ensuring that the most recent copy of the CMS QAP is made available to all personnel involved with the Cemex - Lyons Cement Plant Facility in Lyons, CO.

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1 Introduction

This Quality Assurance Plan (QAP) has been developed for the continuous emissions monitoring systems (CEMS) and continuous opacity monitoring system (COMS) for the Cemex Lyons Plant. The plant operates under Permit #950PBO082 and is within the jurisdiction of USEPA Region 8 and the Colorado Department of Public Health and Environment.

The objective of the QAP manual is to establish a series of activities and documentation that will provide a high level of confidence in the data reported by the CMS and will provide a guideline for the activities that are required to maintain air compliance at the facility.

- **Quality Control (QC)** is defined as the procedures, policies, and corrective actions necessary to ensure product quality. QC procedures are typically routine activities. These activities include but are not limited to daily calibrations and routine maintenance.

- **Quality Assurance (QA)** is defined as the series of checks performed to ensure the QC procedures are functioning properly. The activities include but are not limited to relative accuracy test audits.

This manual has been written to meet the requirements in 40CFR Part 60 and the following definitions are used in this CMS/COMS quality assurance program (QAP).

These definitions are consistent with the ones provided in 40 CFR 63.2 or 60, Appendix B, Performance Specification (PS) 1, 2, 3, 4, 6, or 8:

- **Continuous Monitoring System (CMS)** is a comprehensive term that may include, but is not limited to, continuous emission monitoring systems (CEMS), , continuous parameter monitoring systems (CPMS), or other manual or automatic monitoring that is used for demonstrating compliance with an applicable regulation on a continuous basis as defined by the regulation.

- **Continuous Emission Monitoring System (CEMS)** is the total equipment required for the determination of a gas concentration or emission rate. A continuous monitor is one in which the sample to be analyzed passes the measurement section of the analyzer without interruption. The system consists of the following major components:

  - Sample interface
  - Pollutant analyzer
  - Data recorder

- **Continuous Opacity Monitoring System (COMS)** means a continuous monitoring system that measures the opacity of emissions.

- **Continuous Parameter Monitoring System (CPMS)** means the total equipment

that may be required to meet the data acquisition and availability requirements of this part, used to sample, condition (if applicable), analyze, and provide a record of process or control system parameters.

This document presents the QAP for the Cemex CMS in accordance with 40 CFR 63, Subpart A, Section 8, "Monitoring Requirements, and 40 CFR 60, Appendix F, "Quality Assurance Procedures." These procedures outline the minimum quality assurance (QA) requirements necessary for CMS data submittal to the Colorado Department of Public Health & Environment (CDPHE). Source owners and operators are required to develop and implement a QA program to meet or exceed the minimum requirements found in Section 3.0, "QC Requirements." Table 1-1 shows the minimum QA activities required and the sections of the Cemex CMS QAP that address these QA activities.

Table 1-1 Minimum QA Requirements

Minimum QA Requirement	Source	CEMEX CMS QAP
Calibration	40 CFR 60 Appendix F/ 40 CFR 63 Subpart A	Section 3.1 (CEMS) Section 4.1 (COMS) Section 5.1 (CPMS)
Calibration Drift (CD) determination and adjustment of CMS	40 CFR 60 Appendix F/ 40 CFR 63 Subpart A	Section 3.1.3 (CEMS) Section 4.1.3 (COMS)
Preventive maintenance of CMS	40 CFR 60 Appendix F/ 40 CFR 63 Subpart A	Section 6.2
Data recording, calculations, and reporting	40 CFR 60 Appendix F/ 40 CFR 63 Subpart A	Section 7.0
Accuracy audit procedures	40 CFR 60 Appendix F/ 40 CFR 63 Subpart A	Section 3.2 - 3.3 (CEMS) Section 4.2 – 4.3 (COMS) Section 5.2 (CPMS)
Program of corrective action for malfunctioning CMS	40 CFR 60 Appendix F/ 40 CFR 63 Subpart A	Section 3.4 (CEMS) Section 4.4 (COMS) Section 5.3 (CPMS)

The chapters set forth in this QAP are therefore an integral part in the collection of accurate emissions data. This document is written to the level of expertise and training expected to exist for the various positions responsible for each activity.

This QAP has been developed to assist Cemex personnel in the performance of routine CMS activities such as maintenance and auditing. The chapters in this document provide general system operating and maintenance (O&M) requirements. If needed, specific O&M procedures will be referenced. This document does not provide startup, shutdown, and malfunction (SSM) procedures.

1.1 Section Synopsis

- Section 2 is a brief description of the facility and the CMS.
- Section 3 presents the minimum quality assurance/quality control (QA/QC)

requirements for the CEMS.

- Section 4 presents the minimum quality assurance/quality control (QA/QC) requirements for the COMS.

- Section 5 presents the minimum quality assurance/quality control (QA/QC) requirements for the CPMS.

- Section 6 describes the CMS Maintenance program.

- Section 7 presents information on data recording, calculations, and reporting requirements.

1.2 Acronyms

ACA	Absolute Correlation Audit
acfm	actual cubic feet per minute
CAP	Corrective Action Procedure
CD	Calibration Drift
CDPHE	Colorado Department of Health & Environment
CEMS	Continuous Emission Monitoring System
CFR	Code of Federal Regulations
CGA	Cylinder Gas Audit
CMS	Continuous Monitoring System
CO	Carbon Monoxide
COMS	Continuous Opacity Monitoring System
CPMS	Continuous Parameter Monitoring System
DAHS	Data Acquisition and Handling System
EPA	U.S. Environmental Protection Agency
FID	Flame Ionization Detector
Hg	Mercury
MACT	Maximum Achievable Control Technology
NESHAP	National Emission Standards for Hazardous Air Pollutants
NO	Nitric Oxide
NO_2	Nitrogen Dioxide
NO_x	Nitrogen Oxides
O_2	Oxygen
O&M	Operation and Maintenance
PC	Portland cement
ppm	parts per million
PS	Performance Specification
QA	Quality Assurance
QC	Quality Control
RATA	Relative Accuracy Test Audit
RT	Response Time

scf	standard cubic feet
scfm	standard cubic feet per minute
SO_2	Sulfur Dioxide
SP	Standard Procedure
THC	Total Hydrocarbon
VOC	Volatile Organic Compound
WTA	Walk-Through Audit

2 Facility and CMS Description

This section describes the cement manufacturing process and the associated continuous monitoring systems used for compliance with regulatory requirements.

2.1 Facility Description

Cemex operates a Portland cement facility in Lyons, CO. Limestone and other raw materials are placed on a belt by means of a front-end loader to be processed through a primary crusher, dryer, and a secondary crusher. The material is then stored in silos. Material from the silos is discharged to weigh belts for the formulation of a desired product. The weigh belts discharge to the raw mill. The raw mill mixes and crushes the blended materials and delivers the homogenized material to storage silos. The homogenized material from the silos is delivered to the calciner portion of the kiln. Pulverized coal and natural gas are fired at the bottom of the flash calciner. The calcined material from the calciner then enters the rotary kiln. The material travels towards the clinker discharge end where additional pulverized coal is fired for the clinkering process. The clinker is discharged from the kiln into the clinker cooler where it is cooled by air forced through the clinker bed by undergrate fans. The cooled clinker is then moved to storage stockpiles. The stored clinker is the raw material for the finish mill. In the finish mill the clinker is combined with additives, ground to a fine material, and stored in product silos. The material in the product silos can be loaded for bulk transport or sent to a packaging system.

2.1.1 Organizational Chart

Table 2-1 presents an organizational chart for selected positions at the Cemex facility that shows their respective areas of responsibility in relation to the operation of the CMS.

Table 2-1 CMS Organizational Chart

Position	CMS QA/QC Function
Environmental Manager	CMS QA/QC Coordinator
Environmental Specialist	CMS QA/QC Coordinator (Alternate)
Electrical Supervisor	CMS Operation and Maintenance (O&M) Coordinator

2.1.2 Emission Standards and Applicable Requirements

Applicable requirements for the Cemex plant are detailed in the facility's operating permit and include the Performance Specifications (PS) and QA procedures shown in Table 2-2.

Table 2-2 Applicable Requirements for the CMS

Applicable Requirements
Air Operating Permit
40 CFR 60, Subpart LLL 63.1354(b)(9)-(10) and Subpart F subsumed in 40 CFR 63.1343 - 1350
40 CFR 60, Appendix B:
• PS-1 - Specifications and Test Procedures for Opacity in Stationary Sources
• PS-2 – Specifications and Test Procedures for SO_2 and NO_X CMS in Stationary Sources
• PS-3 – Specifications and Test Procedures for O_2 and CO_2 CMS in Stationary Sources
• PS-4 – Specifications and Test Procedures for CO CMS in Stationary Sources
• PS-6 – Specifications and Test Procedures for Continuous Emission Rate Monitoring Systems in Stationary Sources
• PS-8A – Specifications and Test Procedures for THC Continuous Monitoring Systems in Stationary Sources
• PS-12B – Specifications and Test Procedures for monitoring Total Vapor Phase Mercury Emissions from Stationary Sources using a Sorbent Trap Monitoring System
40 CFR 60, Appendix F: Quality Assurance (QA) Requirements: • Procedure 1 – QA/QC for Gas Analyzers • Procedure 2 – QA/QC for Particulate Monitoring Systems • Procedure 3 – QA/QC for Opacity Monitoring Systems • Procedure 5 – QA/QC for Mercury Monitoring Systems

2.2 CMS Description

The CEMS consists of a multi-component analyzer that measures concentrations of nitrogen oxide (NO), sulfur dioxide (SO_2), carbon monoxide (CO), oxygen (O_2), carbon dioxide (CO_2) and moisture from the Kiln exhaust stack.

In addition, the CEMS configuration includes a mercury monitoring sorbent trap system, a Total Hydrocarbon (THC) analyzer, and a particulate monitor (PM) CPMS at the Kiln stack. A second PM CPMS is installed on the Clinker Cooler Stack and a second THC analyzer is installed on the Raw Material Dryer Stack.

The facility also has a gas flow and opacity monitoring systems on the Kiln stack and an opacity monitor on the Clinker Cooler stack. There are existing dioxin-furan (D/F) temperature transmitters installed on the Kiln and Hart baghouse inlets.

All measurements are done on a real time basis. The system includes a programmable logic controller (PLC). The PLC communicates, via Ethernet, from the CEMS to the Data Acquisition and Handling System (DAHS) computer. The PLC will transmit one-minute averages. Contact closures are provided for alarms and system status.

The CMS components are listed in Table 2-3.

Table 2-3 CMS Configuration

Location	Component	Component Manufacturer/Model
Kiln & Hart Baghouse	Thermocouple	Type J
Kiln & Cooler Stacks	PM Monitor	SICK SP100
Kiln Stack	Volumetric Flow Rate Monitor	TML ULTRAFLOW 150
Kiln & Cooler Stacks	Opacity	LAND 4500 MKII
Kiln Sample System	Sampling Probe(s)	M&C SP 2020
	Heated Sample and Calibration Lines	Thermon ME-2C3-57-1-2C2-ATPVC-047/047-3/12-4/18-TCKX-XINS
CEMS Shelter	NO, SO_2, CO, CO_2, O_2, H_2O Analyzer	SICK MCS100E
	THC Analyzer (Kiln Compliance)	SICK GMS-810 FIDOR
	THC Analyzer (Dryer Compliance vial OHAP's Standard)	SICK GMS-840 FIDOR
	Hg – PS-12B	APEX XC 6000 Sorbent Trap
	Opacity	LAND 4500 MKII
	Programmable Logic Controller (PLC)	Allen Bradley ControlLogix
	Data Acquisition and Handling System (DAHS)	Teledyne Monitor Labs DAHS

3 CEMS QA/QC Program

The QA/QC procedures presented in this section are periodic checks on the CEMS so that any monitoring problems may be identified and corrected. The QA/QC procedures will consist of the following activities to ensure that quality data are obtained on a continuing basis:

- Calibration checks,
- Periodic audits, and
- Corrective action procedures (CAPs).

Tables 3-1 and 3-2 list the CEMS addressed in this section and provides an overview of the QA/QC activities. Each QA/QC activity will be discussed in detail in the following subsections.

Table 3-1 Overview of CEMS QA/QC Activities

CEMS Component	Walk Through Audit	Calibration Drift	Cylinder Gas Audit	Relative Accuracy Test Audit	Corrective Action Plan
NO analyzer	WTA	Daily	Quarterly	Annually	As needed
O_2 analyzer	WTA	Daily	Quarterly	Annually	As needed
SO_2 analyzer	WTA	Daily	Quarterly	Annually	As needed
CO analyzer	WTA	Daily	Quarterly	Annually	As needed
CO_2 analyzer	WTA	Daily	Quarterly	Annually	As needed
THC analyzer	WTA	Daily	Quarterly*	Annually*	As needed
PM CPMS	WTA	Daily	Quarterly**	N/A	As needed
Volumetric Flow Rate monitor	WTA	Daily	Quarterly RAA***	Annually	As needed

Note: Cylinder Gas Audits are performed quarterly except the quarter when a Relative Accuracy Test Audit is conducted.
* THC Analyzer: Quarterly Calibration Error (CE) test per 40 CFR 60 Appendix B, PS-8a. Annual QA RATA required per Appendix B, PS-8.
**Absolute Correlation Audit (ACA) per 40 CFR 60 App, F, Procedure 2
***A quarterly RAA is not performed during the same quarter as the annual RATA.

Table 3-2 Overview of Hg Sorbent Trap QA/QC Activities

CEMS Component	Pre-test Leak Check	Post-test Leak Check	Ratio of Flows	Sorbent Trap Breakthrough	Sorbent Trap Agreement	Spike Recovery
Hg Sorbent Trap	Each Run	Each Run	Each Run	Each Run	Each Run	Each Run

Hg Sorbent Trap quarterly audit (see Table 3-6) and Relative Accuracy (RATA) required for certification and annual QA per PS-12B.

3.1 QA/QC Activities

The purpose of the QA/QC procedures is to provide an assessment of the CEMS reliability and accuracy. The CEMS operational parameters will be checked and adjusted if necessary, during the walk-through audit (WTA) (see Section 3.1.1), while the CEMS accuracy will be assessed by a daily calibration drift (CD) check (see Section 3.1.2).

3.1.1 WTA Procedures

During the WTA, the CEMS Checklist (See Preventative Maintenance Checklist reference located in Appendix A) should be used to verify numerous operational parameters. A "normal" range is given for those items on the checklist that require the operator to read a value on one of the CEMS components and then log it on the Daily CEMS Checklist form.

An explanation of operational parameters to be checked is detailed on the Preventative Maintenance CEMS Checklists. These records are maintained electronically and stored on the facility shared drive. If one of the CEMS operational parameters is not within its desired range, one or more Corrective Actions may be utilized to correct the problem (see Section 3.3). Any malfunctions, adjustments, or corrective actions should be noted on the CEMS Checklist sheet and then placed in the CMS Activity Logbook.

3.1.2 Daily Auto-Calibrations

Per Table 3-3, daily CD checks of the NO, O_2, SO_2, CO, CO_2, and THC analyzers are required by 40 CFR 60, Appendix B and F. The CEMS must check, record, and quantify the Calibration Drift (CD) of each analyzer channel at least once daily. Calibration drift checks are performed approximately 24 hours apart. For NO, O_2, SO_2, CO, and CO_2, tests are run for zero and span (high level): zero level: 0-20% of span and high level: 50-100% of span. For THC, the zero-gas calibration is conducted at a measurement level at or between 0–0.1% of instrument measurement span. The high-level gas calibration is conducted between 50–90% of instrument measurement span.

NO_x is calculated using the measured NO via a conversion factor (1.025) to account for NO_2. The volumetric flow rate monitor also performs a daily CD check to meet calibration requirements of 40 CFR 60, Appendix B, PS 6. The particulate monitor operating as CPMS also performs daily CD tests per 40 CFR 60, Appendix F, Procedure 2, Section 10.2 (1)-(3).

Table 3-3 Calibration Drift (CD) Check Reference Procedures

Parameter	Reference Procedure
NO	40 CFR 60, Appendix B, Performance Specification 2, Section 8.3 (CD)
O_2	40 CFR 60, Appendix B, Performance Specification 3, Section 8.3 of PS-2 (CD)
SO_2	40 CFR 60, Appendix B, Performance Specification 2, Section 8.3 (CD)
CO	40 CFR 60, Appendix B, Performance Specification 4, Section 8.3 of PS-2 (CD)
CO_2	40 CFR 60, Appendix B, Performance Specification 3, Section 8.3 of PS-2 (CD)
THC	40 CFR 60, Appendix B, Performance Specification 8A, Section 6.2

PM CPMS	40 CFR 60, Appendix F, Procedure 2, Section 10.2 (1)-(3), 10.4(1)
Volumetric Flow	40 CFR 60, Appendix B, Performance Specification 6, Section 8.1 (CD)

These procedures describe the methods used to check the response of the instruments to low and high-level calibration standards and then to calculate the resultant CD. The SICK MCS 100E executes a CD sequence each day using a zero gas and gas filter cells for established span values based on reference gases for NO, SO_2, CO and CO_2. The other CEMS analyzers (O_2 and THC) are checked daily for proper function and accuracy by introducing calibration gas to CEMS at the stack probe.

Table 3-4 lists the nominal calibration standard values used for the daily CD check. The calibration values used fall within ranges specified by the PS in 40 CFR 60, Appendix B.

Table 3-4 Zero and Span Ranges for Daily CD Checks

Analyzer/Monitor	Zero Drift Check Range	Span Drift Check Range	Analyzer/Monitor Full Scale
NO	0 ppm	600 – 1,200 ppm	1, 200 ppm
O_2	0%	12.5 - 25%	25%
SO_2	0 ppm	50 - 100 ppm	100 ppm
CO	0 ppm	750-1,500 ppm	1,500 ppm
CO_2	0%	12.5 - 25%	25%
THC (Kiln)	0 ppm	25 - 50 ppm	50 ppm
THC (Dryer) Dual-range	0 ppm	25 – 50 ppm 300 – 600 ppm	600 ppm
PM CPMS (Kiln & Cooler)	0 – 40 mg/m^3	100-200 g/m^3	200 mg/m^3
Volumetric Flow	0 – 80,000 acfm (0" WC)	50-70% of range [(1)]	400,000 scfm

[(1)] As required by 40 CFR §63.1350(n)

The zero and span CD check is performed automatically every morning on the NO, O_2, SO_2, CO, CO_2, PM CPMS and THC analyzers, and the volumetric flow rate monitor. The zero and span CD checks take less than 30 minutes to complete. After the CD check has been completed, the system is allowed to stabilize before logged data will be considered valid again. After the check has been completed, the CD results are then evaluated by the DAHS. The evaluation is a review of the calculated CD for each instrument to determine whether corrective action must be taken. If corrective action is taken, the automated CD check program should be rerun to determine whether the CD is within acceptable limits. The CD check program can be initiated manually utilizing the analyzer panel or DAHS software to determine whether the CD is within acceptable limits.

The results of the daily calibrations not only provide documentation of proper system operation but also give the operator a means of tracking system performance over time so that maintenance scheduling can be optimized. Documentation of the daily CD check is

recorded by the DAHS.

3.1.3 Daily Calibration Drift Evaluation Procedures

The calibration reports are maintained in the DAHS Reports can be viewed for the current and previous days utilizing the DAHS software.

On the SICK MCS-100E, daily zero and span drift checks shall be performed in accordance with the Appendix B and F of 40 CFR 60 using zero gas and internal gas filters. The MCS100E channels are initially calibrated with a certified span gas reference. Each time the MCS100E calibration is adjusted with a span gas reference, an internal filter calibration check is required to establish the new internal filter reference value for the specific gas (e.g. SO2). The MCS100E channels shall be checked with certified span gas when the CD drift error with the internal filter is significant to trigger further action outlined in Table 3-5.

The THC and O_2 analyzers utilize both zero and span certified gas for their daily reference during the CD check routine. Table 3-5 shall be the reference guide for determining the appropriate action to take in the event there is significant drift error between the reference gas the analyzer reading.

The QC requirements of 40 CFR Appendix F require the instrument calibration to be adjusted whenever the daily CD exceeds twice the limits of the applicable PS. Table 3-5 presents a summary of the daily CD limits and the course of action to follow if these limits are exceeded.

As stated in the previous section, the volumetric flow rate monitor also performs daily CD checks of the volumetric flow rate. The daily CD limit in Table 3-5 for volumetric flow rate is derived from 40 CFR 60, Appendix B, PS-6.

Table 3-5 CEMS Calibration Drift Limits

	Case 1		Case 2		Case 3	
	In compliance; action optional (1 X CD), Control Limit		Out of compliance; action required (2 X CD) Out of control if ≥ 5 days		Out of control; immediate action required (4 X CD)	
	Result	Action	Result	Action	Result	Action
NO	±2.5% of span	Adjust calibration if desired	± 5% of span	Adjust calibration	± 10% of span	Adjust calibration
O_2	±0.5% Difference	Adjust calibration if desired	± 1.0% O_2 difference	Adjust calibration	± 2.0% O_2 difference	Adjust calibration
SO_2	±2.5% of span	Adjust calibration if desired	± 5% of span	Adjust calibration	± 10% of span	Adjust calibration
CO	±5% of span	Adjust calibration if desired	± 10% of span	Adjust calibration	± 20% of span	Adjust calibration
CO_2	±0.5% Difference	Adjust calibration if desired	± 1.0% CO_2 difference	Adjust calibration	± 2.0% CO_2 difference	Adjust calibration
THC	±3% of span	Adjust calibration	± 6% of span	Adjust	± 12% of span	Adjust

		if desired		calibration		calibration
PM CPMS	±2% Difference	Adjust calibration if desired	± 4% PM difference	Adjust calibration	± 8% PM difference	Adjust calibration
Volumetric Flow Rate	±3% of span	Adjust calibration if desired	± 6% of span	Adjust calibration	± 12%of span	Adjust calibration

Note: The absolute value of the CD results should be used to determine the appropriate action to take.

In Case 1, no corrective action is required since the CEMS instruments are in compliance with the PS found in 40 CFR 60, Appendices B and F and this QA Plan.

However, corrective action can be taken if desired and should be taken as the indicated limit is approached. If corrective action is taken, it is a preventive measure in order to lessen the likelihood that a future CEMS CD check may be “out of compliance” or “out of control.” If manual calibration and/or repair of the offending instrument are performed, any corrective action taken should be recorded in the CMS Activity Logbook (see Section 3.3 to determine what CAP should be taken for corrective action, subsequently referred to as CA). Normal operation is resumed following completion of a successful CD check.

In Case 2, the instrument is deemed “out of compliance.” If this CD occurs for five consecutive daily periods, the instrument is considered “out of control” and therefore corrective action must be taken immediately. The DAHS alarm system, supported by inspection of the daily calibration report, will alert personnel if a CD exceedance has occurred. Manual calibration and/or repair of the offending analyzer should be performed, and any corrective action taken recorded in the CMS Activity Logbook (see Section 3.3 to determine what CAP should be taken). Normal operation is resumed following completion of a successful CD check.

In Case 3, the instrument is deemed “out of control.” based on the results of a single CD check. In this situation, corrective action must be taken immediately. The DAHS alarm system, supported by inspection of the daily calibration report, will alert personnel when an analyzer is classified as “out of control” by the DAHS. Corrective action must be taken immediately since the CMS data collected during a period in which the CD results are “out of control” are flagged as invalid and therefore cannot be used for compliance purposes. Out of control status flags data “invalid” back to the last passing CD test. Manual calibration and/or repair of the offending analyzer are to be performed and corrective action recorded in the CMS Activity Logbook (see Section 3.3 to determine what CAP should be taken). Normal operation is resumed following completion of a successful CD check.

To ensure compliance during any 24-hour period, Cemex Instrument Technicians are available. For instance, if a failed calibration occurs on a weekend, personnel will be available to perform corrective action and to conduct a passed calibration. This will reduce reporting of downtime during an excursion event.

Span gas calibration is typically performed only if internal span calibrations fail. The following is a brief description of full manual gas calibration performed on a SICK MCS-100E when needed, e.g., if gas filter daily internal calibrations fail.

First, a zero gas is run through the system and allowed adequate time to stabilize. The analyzer reading is then recorded and compared to the criteria set in the federal regulations. If the analyzer response is outside of the criteria, the zero point is adjusted to zero, and this new zero point is recorded. Then a certified span gas is run through the system until stabilization is met, and the same procedure is repeated for the span setting of the analyzer. If the calibration is within the required calibration error criteria, the analyzer is acceptable for use. In the event that the analyzer fails internal gas filter calibration checks, the analyzer may need corrective maintenance, or the gas filters may need to be recertified.

3.1.4 Mercury Sorbent Trap Batch Checks

Batch samples are taken at pre-determined time intervals for the sorbent trap mercury monitoring system. Each batch sampling period must meet the required QA/QC checks as outlined in the following Table 3-6. Refer to the manufacturer's manual for information on setting up test runs and trap checks, and for detailed QA procedures.

Table 3-6 Mercury Sorbent Trap Quality Assurance – PS-12B

QA/QC Test or Specification	Acceptance Criteria	Frequency	Consequences if not met
Pre-test leak check	≤4% of target sampling rate	Prior to starting monitoring period	Monitoring must not commence until the leak check is passed.
Post-test leak check	≤4% of average sampling rate	After monitoring period is completed	Invalidate the data from the paired traps or, if certain conditions are met, report adjusted data from a single trap (see PS-12B, Section 12.8.3).
Ratio of stack gas flow rate to sample flow rate	Maintain within 5% of the hourly ratios or 5 hourly ratios (whichever is less restrictive) may deviated from the reference ratio by more than ±25%	Every hour throughout monitoring period.	Invalidate the data from the paired traps or, if certain conditions are met, report adjusted data from a single trap (see PS-12B, Section 12.8.3).
Sorbent trap section 2 breakthrough	≤5% of Section 1 Hg mass ≤10% of Section 1 Hg mass if average Hg concentration is ≤µg/scm No criterion when Hg concentration for trap is <10% of the applicable emission limit (must meet all other QA/QC specifications)	Every sample	Invalidate the data from the paired traps or, if certain conditions are met, report adjusted data from a single trap (see PS-12B, Section 12.8.3).

Paired sorbent trap agreement	≤10% Relative Deviation (RD) if the average concentration is >1.0 µg/m^3 ≤20% RD if the average concentration is ≤1.0 µg/m^3 Results also acceptable if absolute difference between concentrations from paired traps is ≤0.03 µg/m^3	Every Sample	Either invalidate the data from the paired traps or report the results from the trap with the higher Hg concentration.
Spike recovery study	Average recovery between 85% and 115% for each of the 3 spike concentration levels	Prior to analyzing field samples and prior to use of new sorbent media	Field samples must not be analyzed until the percent recovery criteria have been met.
Multipoint analyzer calibration	Each analyzer reading within ±10% of true value and $r^2 \geq 0.99$	On the day of analysis, before analyzing any samples	Recalibrate until successful.
Analysis of independent calibration standard	Within ±10% of true value	Following daily calibration, prior to analyzing filed samples.	Recalibrate and repeat independent standard analysis until successful.
Spike recovery from section 3 of both sorbent traps	75-125% of spike amount	Every sample	Invalidate the data from the paired traps or, if certain conditions are met, report adjusted data from a single trap (see PS-12B, Section 12.8.3).
Relative Accuracy	RA ≤20% of RM mean value; or if RM mean value ≤5.0 µg/scm, absolute difference between RM and sorbent trap monitoring system mean values ≤1.0 µg/scm	For initial certification and annually thereafter.	Data from the system is invalid until a RATA is passed.
Gas flow meter calibration	An initial calibration factor (Y) has been determined at 3 settings; for mass flow meters, initial calibration with stack gas has been performed. For subsequent calibrations, Y within ±5% of average value from the most recent 3-point calibration	At 3 settings prior to initial use and at least quarterly at one setting thereafter	Recalibrate meter at 3 settings to determine a new value of Y.
Temperature sensor calibration	Absolute temperature measured by sensor within ±1.5% of a reference sensor	Prior to initial use and at least quarterly thereafter	Recalibrate; sensor may not be used until specification is met.

Barometer calibration	Absolute pressure measured by instrument within ±10 mm Hg of reading with a NIST-traceable barometer	Prior to initial use and at least quarterly thereafter	Recalibrate; instrument may not be used until specification is met.

Note: If both traps fail to meet the acceptance criteria, the data from the pair of traps are invalidated. However, if only one of the paired traps fails to meet this particular acceptance criterion and the other trap meets all of the applicable QA criteria, the results of the valid trap may be used for reporting provided that the measured Mercury concentration is multiplied by a factor of 1.111. The last three rows of table 3-6 above comprise the Mercury Sorbent Trap System **Quarterly Audit procedure**.

The APEX sorbent trap mercury system has an Alarm Actions utility screen. Each alarm condition has several parameters that may be set. For each sensor input on the Alarm Actions screen, a valid range or upper limit may be set, depending on sensor type. Most alarms feature an auto-reset function in the case of a non-critical alarm condition.

3.2 Quarterly QA/QC Activities

40 CFR 60, Appendix F requires that each CEMS analyzer be audited at least each calendar quarter and that successive quarterly audits shall occur no closer than 2 months. The audit procedures are described in 40 CFR 60, Appendix F: which include: relative accuracy test audits (RATAs) and cylinder gas audits (CGAs). CGAs are conducted quarterly except when a RATA must be conducted annually for one of the four-calendar quarters. CGAs and RATAs are described in the following sections.

Quarterly relative accuracy audits (RAAs) and an annual RATA is performed per 40 CFR 60, Appendix B, PS-6 for volumetric flow rate monitors. When Cemex emission limits are based on mass emission rates, the accuracy of the volumetric flow rate is critical.

A quarterly linearity or absolute correlation audits (ACAs) per 40 CFR 60 App. F, Procedure 2 Section 10.3 (1)-(2) against Section 10.4(3) criteria is conducted for the PM CPMS. An ACA is performed quarterly using three neutral density filters. Each filter will be in place for 3 minutes, and the values recorded. These filters will be installed three times and the average of the runs will be verified against the known values of the filters.

3.2.1 Cylinder Gas Audit (CGA)

40 CFR 60, Appendix F requires a CGA to be performed in three of four calendar quarters. CGAs may not be conducted in more than three quarters in succession, and any two successive quarterly audits shall occur no closer than 2 months apart. The purpose of the CGAs are to provide a more rigorous and independent assessment of the CEMS accuracy and reliability than is provided by the daily calibrations. A CGA is not required for the volumetric flow rate monitor.

A CGA for the extractive CEMS analyzers will be performed once each quarter except when a RATA is to be performed. During a CGA, the CEMS will be challenged at two points within the concentration ranges depicted in Table 3-7. The CGA will be performed in accordance with 40 CFR 60, Appendix F, Procedure 1, Section 5.1.2.

Table 3-7 CGA Calibration Gas Concentration Ranges

Audit Point	Audit Gas Range			
	NO, SO_2, CO Analyzers	THC Analyzer	O_2 Analyzer	CO_2 Analyzer
1	20% to 30% of span value	0 – 0.1% of span	4% to 6%	5% to 8%
2	50% to 60% of span value	30 – 40% of span	8% to 12%	10% to 14%
3	N/A	70 80% of range	N/A	N/A

THC Analyzer: Quarterly Calibration Error (CE) test per 40 CFR 60 Appendix B, PS-8A.

The general procedure for performing a CGA is to introduce calibration gas standards to the extractive CEMS analyzers at the sample probe. The response of the analyzer is then compared to the known concentration of the gas in the cylinder. The CEMS must be challenged three times at each audit point, and the average of the three responses will be used to determine accuracy. Details for performing CGA activities are located in the instrument technical manuals.

To successfully pass the CGA, the difference between the CEMS response and the audit value must be ±15% of the average audit value or ±5 ppm. For diluent monitors, the difference between the CEMS response and the audit value must be ±15% of the average audit value. For THC, the difference between the CEMS and reference values must be no greater than ±5% of the span value. If these criteria are exceeded, the analyzer will be designated as “out of control” and corrective action should be taken to identify and fix the problem.

After the corrective action has been performed, another CGA will be conducted on the unit to verify that the CEMS has returned to good working order and is operating within specifications.

During the period that the analyzer is “out of control,” data from that analyzer will not be used to calculate emission compliance. The “out of control” period ends after adjustments or corrective action has been taken and the latest CGA test results are acceptable.

If the CEMS repeatedly fails the CGA, it will be necessary to conduct comprehensive diagnostic testing and troubleshooting of the analyzer in question using the procedures described in the manufacturers’ owner’s manual. The procedures in this QAP will also be reviewed to address excessive inaccuracies and revise the plan if necessary.

3.3 Annual QA/QC Activities

Upon installation of the CEMS for initial certification and once a year thereafter, a RATA will be conducted for the CEMS analyzers. The RATA will be performed in accordance with the procedures in 40 CFR 60, Appendix F.

The purpose of the annual RATA is to provide a more rigorous and independent QA assessment of the CEMS accuracy and reliability than is provided by the quarterly CGA. Procedures for RATA testing are maintained by the outside contractor conducting the test.

Please note: NESHAP Subpart LLL requires an annual THC RATA per PS-8 for quality assurance. There is also quarterly calibration error testing. Although certified PS-8A, the THC CEMS will require certification RATA testing per PS-8. The PM CPMS is not subject to PS-11 or RATA testing, but annual PM compliance testing is required to establish the PM CPMS monitoring ranges.

The RATA will be performed in accordance with the most recently approved CEMS RATA Test Notification Protocol. The EPA Reference Methods that will be utilized to conduct the RATA testing are presented in Table 3-8.

Table 3-8 RATA Reference Methods

RATA	Reference Methods	Parameter
NO_x	40 CFR 60, EPA Reference Method 7E	NO_x Concentration
O_2	40 CFR 60, EPA Reference Method 3A	O_2 Concentration
SO_2	40 CFR 60, EPA Reference Method 6C	SO_2 Concentration
CO	40 CFR 60, EPA Reference Method 10	CO Concentration
CO_2	40 CFR 60, EPA Reference Method 3A	CO_2 Concentration
THC	40 CFR 60, EPA Reference Method 25A	THC Concentration
Hg Sorbent Trap	40 CFR 60, EPA Reference Method 30B	Hg Concentration
Volumetric Flow Rate	40 CFR 60, EPA Reference Methods 1 & 2	Velocity & Flow Rate
Moisture	40 CFR 60, EPA Reference Method 4	Moisture % (no RA)

Note: Relative accuracy will be assessed on a lb/hour basis when average emissions during the test are < 50% of limit, which requires concentrations and volumetric flow rates. The RA of the individual quantities may be assessed.

Based on the RATA results, the CEMS will either be “in compliance” or be “out of control” and require corrective action. Table 3-9 shows the criteria used to make this determination.

Table 3-9 Criteria for Passing a RATA

Instrument	Procedure	Acceptable Performance Criteria
NO_x Analyzer	RATA	20% RA (10% RA)
O_2 Analyzer	RATA	20% RA or ≤1.0% of actual O_2 Concentration
SO_2 Analyzer	RATA	20% RA (10% RA)
CO Analyzer	RATA	10% RA (5% RA)
CO_2 Analyzer	RATA	20% RA or ≤1.0% of actual CO_2 Concentration
THC Analyzer	RATA	20% RA (10% RA) *
Hg Sorbent Trap	RATA	20% RA**
Volumetric Flow Rate Monitor	RATA	10% RA***

Note: Values in parenthesis are criteria if applicable standard is used to calculate relative accuracy.
*Although PS-8A, THC annual RATA is required per PS-8.
**Otherwise, if Hg RM data is ≤5.0 ug/m3, then 1.0 ug/m3 difference.
***10% flow RA applicable per NESHAP, 40 CFR 63.1350(n)(8)(i,ii)

RATA performance criteria are determined by the concentration of the criteria pollutants and mercury emitted from the source. If the criteria pollutant is greater than 50% of the site permit limit, then the RATA performance criteria is as listed in Table 3-9. If the criteria pollutant is less than 50% of the site permit limit, then the RATA performance criteria is listed in parenthesis in Table 3-9. THC is acceptable if relative accuracy does not exceed 20% of the mean value of the RM test data in terms of units of the emission standard or 10% of the applicable standard (permit limit), whichever is greater.

If the CEMS exceeds the RATA criteria and therefore is deemed “out of control,” corrective action will be taken on the analyzer/monitor of concern to identify and fix the problem. After the corrective action has been performed, another RATA will be conducted on the unit to verify that the CEMS has returned to good working order and is operating within specifications.

During the period that the analyzer/monitor is “out of control,” data from that instrument will not be used to calculate emission compliance. The “out of control” period ends after adjustments or corrective action has been taken and RATA test results are acceptable. Details for conducting RAA activities are the responsibility of an outside contractor performing the RATA. Once performed, RATA records are kept by the plant.

3.4 Corrective Action Procedures (CAPs)

Corrective Actions Procedures (CAPs) will be used when the CEMS performance is unacceptable or “out of control,” as indicated by either a daily, quarterly, or annual QA/QC check or a CEMS alarm. If one of the CEMS operational parameters is not within its desired range, CAP guidelines may be utilized to correct the problem.

The purpose of this section is not to reproduce the maintenance and repair procedures provided in the technical manuals for each component, but to provide a general guideline of the procedures used to perform CAPs and documenting any action taken.

3.4.1 CAPs for CEMS

The most common corrective action (CA) to be performed will be a manual adjustment of an analyzer during a manual CD check using calibration gases. The CD check may be initiated for a single analyzer (for any offending instrument channels) from the CEMS control panel. Alternatively, the CEMS calibration program may also be initiated through the DAHS. Manual adjustments of an analyzer should not be performed during the daily automated CD check. Care should be taken to wait an appropriate time after the calibration gas flow is initiated for the analyzer response to stabilize before performing a manual adjustment on any of the analyzers or analyzer instrument channels.

If the CMS does not respond to manual adjustments, Cemex will contact the CEMS contractor to make the appropriate repairs to the CEMS. After the CMS has been repaired, the repair needs to be documented in the CMS Activity Logbook and the CEMS should be appropriately tested to demonstrate it still meets the applicable standards.

4 COMS QA/QC PROGRAM

The QA/QC procedures presented in this section are periodic checks on the COMS so that any monitoring problems may be identified and corrected. The QA/QC procedures will consist of the following activities to ensure that quality data are obtained on a continuing basis:

- Calibration checks,
- Periodic audits and Corrective action procedures (CAPs) guidelines.

Table 4-1 lists the COMS addressed in this section and provides an overview of the QA/QC activities. Each QA/QC activity will be discussed in detail in the following subsections.

Table 4-1 Overview of COMS QA/QC Activities

COMS Component	Walk Through Audit	Calibration Drift	Performance Audit	Zero Alignment	Corrective Action Plan
Opacity monitor (Main Kiln Stack)	WTA	Daily	Quarterly	Annually	As needed
Opacity monitor (Clinker Cooler Stack)	WTA	Daily	Quarterly	Annually	As needed

4.1 QA/QC Activities

The purpose of the QA/QC procedures is to provide an assessment of the COMS reliability and accuracy. The COMS operational parameters will be checked and adjusted if necessary, during the walk-through audit (WTA) (see Section 4.1.1); while the COMS accuracy will be assessed by a daily calibration drift (CD) check (see Section 4.1.2).

4.1.1 WTA Procedures

During the WTA, the COMS Checklist (See the Preventative Maintenance Checklist reference in Appendix A) should be used to verify numerous operational parameters. A “normal” range is given for those items on the checklist that require the operator to read a value on one of the CMS components and then log it on the COMS Checklist form.

An explanation of the operational parameters to be checked is detailed on the Daily COMS Checklist. The Daily COMS Checklist is maintained electronically. If one of the COMS operational parameters is not within its desired range, one or more Corrective Actions may be utilized to correct the problem (see Section 4.3). Any malfunctions, adjustments, or corrective actions should be noted on the Daily COMS Checklist and then placed in the CMS Activity Logbook.

4.1.2 Daily Auto-Calibrations

Procedure 3 for COMS in Appendix F and PS-1 requires daily CD check for opacity. These requirements are included in this QAP. The reference procedure to check the zero and span CD of the COMS is listed in Table 4-2.

Table 4-2 Calibration Drift Check Reference Procedures

Parameter	Reference Procedure
Opacity	40 CFR 60, Appendix B, Performance Specification 1, Section 8.4 (CD)

The procedure describes the methods used to check the response of the instruments to zero and upscale optical standards and then to calculate the resultant CD. By introducing the optical standards to the COMS, the monitor is checked for proper function and accuracy.

The zero and upscale CD check is performed automatically every morning on the opacity monitor. The zero and upscale CD checks take less than 15 minutes to complete. After the CD check has been completed, the system is allowed to stabilize before logged data will be considered valid again. After the check has been completed, the CD results are then evaluated by the DAHS according to reference procedures outlined in 40 CFR 60, Appendix B, PS-1, Section 8.4. The evaluation is a review of the calculated CD for each instrument to determine whether corrective action must be taken. If corrective action is taken, the automated CD check program should be rerun to determine whether the CD is within acceptable limits. The CD check program can be initiated manually on the instrument remote panel or by utilizing the DAHS software to determine whether the CD is within acceptable limits.

The results of the daily calibrations not only provide documentation of proper system operation but also give the operator a means of tracking system performance over time so that maintenance scheduling can be optimized. Documentation of the daily CD check is recorded by the DAHS.

4.1.3 Daily Calibration Drift Evaluation Procedures

The calibration reports are maintained in the DAHS. Reports can be viewed for the current and previous days utilizing the DAHS software.

Table 4-3 presents a summary of the daily CD limits and the course of action to follow if these limits are exceeded.

Table 4-3 COMS Calibration Drift Limits

Source	Case 1		Case 2	
	In Compliance; Action Optional (1 x CD), Control Limit		Out of Control; Action Required (2 x CD)	
	Result	Action	Result	Action
Opacity (Main Kiln Stack)	±2.0% opacity	Adjust Calibration if Desired	±4.0% opacity	Adjust Calibration
Opacity (Clinker Cooler Stack)	±2.0% opacity	Adjust Calibration if Desired	±4.0% opacity	Adjust Calibration

Note: The absolute value of the CD results should be used to determine the appropriate action to take.

In Case 1, no corrective action is required since the COMS instruments are in compliance with PS-1 and Procedure 3 criteria found in 40 CFR 60, Appendices B and F and in this QA Plan. However, corrective action can be taken if desired and should be taken as the indicated limit is approached. If corrective action is taken, it is a preventive measure in order to lessen the likelihood that a future COMS CD check may be "out of compliance" or "out of control." If manual calibration and/or repair of the offending instrument are performed, any corrective action taken should be recorded in the CMS Activity Logbook (see Section 4.3 to determine what CAP should be taken). Normal operation is resumed following completion of a successful CD check.

In Case 2, the instrument is deemed "out of control" and out of compliance based on the results of a single CD check. In this situation, corrective action must be taken immediately. The DAHS alarm system, supported by inspection of the daily calibration report, will alert personnel when an analyzer is classified as "out of control" by the DAHS. Corrective action must be taken immediately since the COMS data collected during a period in which the CD results are "out of control" are flagged as invalid and therefore cannot be used for compliance purposes. Manual calibration and/or repair of the offending analyzer are to be performed and corrective action recorded in the CMS Activity Logbook (see Section 4.3 to determine what CAP should be taken). Normal operation is resumed following completion of a successful CD check.

To ensure compliance during any 24-hour period, Cemex Instrument Technicians are available 24 hours a day, seven days a week, 52 weeks a year.

4.2 Quarterly QA/QC Activities

There are three quarterly performance auditing requirements outlined in the 40 CFR 60, Appendix F, QA/QC Procedure 3 for COMS. The requirements are:

- Zero Compensation – If the COMS has an automatic zero compensation function,

then the value of the zero-compensation applied at the time of the audit must calculated as equivalent opacity, corrected to stack conditions, according to procedures specified by the manufacturer. This procedure, called a “Dust Correction” can be conducted with lens cleaning as needed for drift. The zero compensation must be less than 4%.

- Calibration Error Test – A three-point calibration error (CE) test must be conducted using three neutral density filters. These Quarterly COMS Audit procedures for conducting a CE test are contained in 40 CFR 60, Appendix B, PS-1. The CE must be less than 3%.

- Check Optical Alignment – The optical alignment must be checked when the stack temperature is ±20% of the typical operating temperature, in °F. The optical alignment must be less than 3%.

If the COMS exceeds one or more of the criteria for the quarterly performance audit, it indicates the COMS is not in compliance and out-of-control. The out-of-control period begins at the completion of a performance audit indicating an unacceptable performance. The out-of-control period ends when the completion of the corrective action(s) and subsequent successful audit. Details for the COMS QA/QC activities are provided in instrument technical manuals.

4.3 Annual QA/QC Activities

An annual requirement in the 40 CFR 60, Appendix F QA/QC Procedure 3 for COMS is a zero path alignment. The zero path alignment must be conducted under clear path conditions. Procedure 3 requires the annual alignment test be conducted “off-stack.” The COMS must be removed from the stack, so the monitor path is free of PM.

The zero alignment must be less than 3%, otherwise the COMS is considered out of control.

4.4 Corrective Action Procedures (CAPs)

Corrective Actions Procedures (CAPs) will be used when the COMS performance is unacceptable or “out of control,” as indicated by either a daily, quarterly, or annual QA/QC check or a COMS alarm. If one of the COMS operational parameters is not within its desired range, CA guidelines may be utilized to correct the problem.

The purpose of this section is not to reproduce the maintenance and repair procedures provided in the technical manuals for each component, but to provide a general guideline of the procedures used to perform CAs and documenting any action taken.

4.4.1 CAPs for COMS

The most common CA to be performed will be a manual adjustment of the COMS during a CD check. The CD check may be initiated for a single analyzer from the COMS control panel. Alternatively, the CMS calibration program may be initiated through the DAHS.

Manual adjustments of an analyzer should not be performed during the daily automated CD check.

If the COMS do not respond to manual adjustments, Cemex will contact the COMS contractor to make the appropriate repairs to the COMS. After the COMS has been repaired, the repair needs to be documented in the CMS Activity Logbook and the CMS should be appropriately tested to demonstrate it still meets the applicable standards.

5 CPMS QA/QC Program

The QA/QC procedures presented in this section are periodic checks on the CPMS so that any monitoring problems may be identified and corrected. The QA/QC procedures will consist of the following activities to ensure that quality data are obtained on a continuing basis:

- Calibration checks,
- Periodic audits and Corrective action procedures (CAPs).

Table 5-1 lists the CPMS addressed in this section and provides an overview of the QA/QC activities. Each QA/QC activity will be discussed in detail in the following subsections.

Table 5-1 Overview of CPMS QA/QC Activities

CPMS Component	Walk Through Audit	Calibration Check	Performance Audit	Corrective Action Plan
Thermocouple	WTA	Daily	Every 3 Months	As Needed

The SO_2 CEMS is utilized as a parametric indication for HCl compliance as well as for compliance with permitted emission limits. The QA/QC program components (daily, quarterly, and annual) for the SO2 CEMS are described in Section 3.0 and are not repeated here. Also, the PM CPMS is utilized as a parametric monitor indicating PM compliance with an output parameter correlated to PM emissions. The QA/QC program components for the PM CPMS are described in Section 3.0 as well.

5.1 QA/QC Activities

The purpose of the QA/QC procedures is to provide an assessment of the CPMS reliability and accuracy. The CPMS operational parameters will be checked and adjusted if necessary, during the WTA.

During the WTA, general guidelines outlined below should be used to verify numerous operational parameters. A “normal” range is given for those items and the operator can read a value on one of the CPMS components and then log it on the CEMS Checklist form.

An explanation of the operational parameters to be checked is detailed on the Daily CEMS Checklist. The Daily CEMS Checklist is maintained electronically. If one of the CMS operational parameters is not within its desired range, one or more Corrective Actions may be utilized to correct the problem (see Section 3.3). Any malfunctions, adjustments, or corrective actions should be noted and then placed in the CMS Activity Logbook.

5.2 QA/QC Activities

Calibration of the Kiln and Hart baghouse inlet thermocouple-potentiometer system will be conducted, at a minimum, once every three months. The calibration will be performed with a National Institute of Standards and Technology (NIST) traceable calibration device. A sample of the form used for this calibration is attached in Appendix C.

5.3 Corrective Action Procedures (CAPs)

Corrective Actions Procedures (CAPs) will be used when the CPMS performance is unacceptable or "out of control," as indicated by either QA/QC check or a CPMS alarm.

If one of the CPMS operational parameters is not within its desired range, one or more CA guidelines may be utilized to correct the problem. The purpose of this section is not to reproduce the maintenance and repair procedures provided in technical manuals for each component but provide a general guideline of the procedures used to perform CAs and documenting any action taken.

5.3.1 CAPs for CPMS

The most common reason for corrective action on baghouse thermocouples will be when thermocouple performance is unacceptable or "out of control," as indicated by either QA/QC check or a CMS alarm. The thermocouples will be replaced if any of the following occur:

- One of the CMS operational parameters is not within its desired range of 0 to 5 degrees higher or lower than the control room during a calibration,
- The control room is reading higher/lower than baghouse handheld calibrator, or
- When an I/O alarm is sent to the control room, meaning a thermocouple has gone bad.

6 CMS Maintenance Program

Maintenance will be included as part of the CMS standard procedures to keep the system components in good working order. In order to reduce the amount of time that the CMS is off-line in the event that a component of the CMS needs to be replaced, redundancy was designed into the CMS for certain critical components. In addition to the redundant analyzers, pumps, and gas conditioners, a limited amount of consumable spare parts will be maintained.

6.1 CMS Log Keeping Procedure

The active CMS Activity Logbook is maintained electronically and stored on the facility shared drive. The following should be included for all log entries:

- Date: month, day, and year.
- Time: military time.
- Event: describe the reason for the entry and any action; and
- Personnel: name of individual making entry.

Examples of maintenance activities that should be recorded include:

- Replacement of calibration gas standards.
- CMS component out-of-service or returned to service.
- Maintenance activities performed (preventive or forced including
- calibration adjustments to the CMS).
- Audit activities (quarterly, annual, other).
- Corrective actions for an out-of-control instrument.

6.2 Preventative Maintenance

In the event that preventative maintenance is required between CGA audits the respective owner's manual for each CMS component should be referenced for specific maintenance or repair activities.

6.2.1 Weekly Maintenance

The purpose of the weekly maintenance is to provide a daily assessment of the CMS reliability and accuracy. The CMS operational parameters will be checked and adjusted if necessary, during the weekly WTA (see Section 3.1.1), while the CMS accuracy will be assessed by a weekly CD check (see Section 3.1.2). The weekly CEMS Checklist found in Appendix A outlines weekly activities and can be used as a log form.

6.2.2 Periodic Maintenance

Preventative maintenance activities are managed through a work order system that is maintained by the Maintenance Department. Work orders are entered for routine preventative maintenance activities on a weekly, monthly, quarterly, and annual recurring basis. An example work order is provided in Appendix A. CMS accuracy is maintained by the daily calibrations, quarterly CGAs, ACAs and RAAs and annual RATAs.

6.3 Spare Parts Inventory

The purpose of the spare parts inventory is to have on site the parts that typically wear out due to usage, such as a pump diaphragm. In addition to spare parts, a supply of consumable parts (e.g., filters) will be maintained. Any parts beyond the manufacturer's standard list of spare parts will be supplied and/or installed by a contracted vendor. See Appendix D for details on spare parts inventory list.

6.4 Maintaining CMS Validity

The requirement to conduct testing after non-routine maintenance or replacement of portions of the CMS is not specifically addressed in regulatory language or guidance documents. It is recognized that performing some types of maintenance on the CMS should be followed by additional QA/QC procedures to ensure that the performance of the system after maintenance still meets acceptable standards. It is also recognized that this testing should be based on the standard audit procedures outlined in 40 CFR 60, Appendix F and may include a daily CD check, and response time (RT) test, a CGA or RAA, and, if necessary, a RATA. The following approach was developed to specify what additional testing will be completed after maintenance is performed.

Typical maintenance activities have been broken down into four categories, which are defined below. These definitions may not include all the potential maintenance situations that might be encountered during operation of the CMS but should cover most circumstances.

- Cleaning and Routine Maintenance - This includes routine maintenance activities such as replacing filters, changing gas bottles, removing dirt/dust buildup, adjusting zero/span settings, or other non-invasive procedures where disassembly of components is not required.

- Minor Repair - No Parts Replaced - This includes periodic maintenance where short-duration downtime (one shift) and some component disassembly is required but reassembly uses no replacement parts. This would include such things as removing condensation or particulate buildup from inside lines or other components, pulling the probe from the duct for any reason, fixing minor leaks, and repairing loose or broken wiring. This would also include adjusting DAHS signal transmitters.

- Major Repair - Identical Replacement Parts - This includes maintenance activities that

involve a component failure, requiring that the component either be removed from service for an extended time or replaced altogether if it is damaged beyond repair. This would include repair or replacement of analyzer detectors, system heaters (filter, sample line, etc.), pumps, sample probe, sample distribution tubing, filter housings, and any DAHS component used to transmit, convert, or process analog or digital information. This would also include situations where the gas sample delivery system is reconfigured or modified.

- Major Repair - Complete or Non-Identical Replacement Parts - This includes maintenance activities that involve a component failure or upgrade where the original component was replaced with anything other than same-brand, factory original, and replacement equipment. This would include the sample probe (including its location), heated sample lines, sample conditioner, pumps, and analyzers. This also would include replacement of the data acquisition hardware or software.

This does not include support items not directly associated with sample acquisition and analysis, such as the condensate removal pump or system wiring (external to the analyzers). It also does not include manufacturer-supplied revisions or upgrades to the DAHS or utilitarian software programming changes. Given this set of maintenance categories and considering the major CMS components, Table 6-1 lists the maintenance category and the system component and identifies the testing that will be completed following the maintenance activity.

Table 6-1 Testing to be Completed Following CMS Maintenance

Component	Maintenance Category			
	Cleaning and Routine Maintenance	Minor Repair – No Parts Replaced	Major Repair – Identical Replacement Parts	Major Repair – Complete or Non-identical Replacement Parts
Heated Sample Probe	No action	CD check	CD check / RT	RATA
Heated Sample Line	No action	CD check	CD check / RT	CGA / RT
Heated Sample Gas Pump	No action	CD check	CD check / RT	CGA / RT
Sample Gas Distribution System	CD check	CD check	CD check / RT	CGA / RT
Gas Analyzer	CD Check	CD check	CGA	RATA
Sorbent Trap Hg System	Leak check	System test	System test	RATA
PM CPMS	CD Check	CD check	CD check	ACA
DAS	No action	No action	No action	CD check
Thermocouple	No action	Calibration	Calibration	Calibration
Volumetric Flow Monitor	CD check	CD check	RATA	RATA

7 Data Recording, Calculations, and Reporting

Data recording, calculations, and reporting will be performed in accordance with 40 CFR 60 Appendix F, and the operating permit. The DAHS will manage data recording and calculations and help facilitate the reporting process.

7.1 Data Recording

Data acquisition and recording is managed by the DAHS, which is capable of reading all values over the full range of the instruments on a prevailing time basis. As required by 40 CFR 60, emission data generated by an “out-of-control” analyzer will be flagged as such and will not be used to calculate emissions.

The DAHS will maintain a chronological file of the following information:

- All 1-minute average data readings from the gas analyzers and the volumetric flow monitor,
- All valid data averages,
- The time period and magnitude of emissions that exceed the permit limits,
- Data and results for all daily calibration checks, and
- The data used to convert concentration data to mass emission data; and
- The time period and cause for any invalid data averages.

7.2 Data Calculations

The data logging component of the DAHS continuously receives inputs from the various analyzers, monitors, and the other CMS instruments. Each minute the data logger calculates a 1-minute average for each data input channel. The data logger also uses the 1-minute readings to calculate other averages. The DAHS software polls the data logger to obtain the calculated 1-minute and other averages and records them in the DAHS computer.

7.2.1 Data Reduction

The following data validation criteria will be used in performing data calculations. Data must be considered invalid if any of the following conditions occur:

- The monitoring system is not operated in accordance with the QA criteria in 40 CFR 60, Appendix F.

- Any portion of the monitoring system is inoperative unless a backup instrument is online.

- The measurement device is considered out-of-control. Data are considered invalid until corrective action is taken, and a CD check is successful.

- Calibration or zero drift check for a measurement device is not conducted during a

calendar day. Data for the following day are considered invalid until a successful CD check is initiated.

- The monitored process is not operating and/or not emitting the pollutant being monitored.

- The required CGA for any measurement device exceeds the applicable performance standard. Data are invalid until the next acceptable CGA.

- The required CGA is not conducted during a calendar quarter. Data for the following quarter are considered invalid until a successful CGA is initiated.

- The system fails a RATA. Data are considered invalid from the time of the failed test until the start of the next acceptable RATA.

- A RATA is not conducted during a calendar year. Data following the end of the calendar quarter during which the annual RATA is required are considered invalid until a successful RATA is started.

All data averages must be calculated using valid data only. The following data reduction criteria apply:

- A 1-minute average will be considered valid if it contains at least 75% valid data readings.

- An hourly average will be considered valid if it contains four valid 15-minute averages (at least one data point in all four quadrants).

- A partial hour when quality assurance is performed will be considered valid if it contains two valid 15-min averages (data points in two quadrants).

- An hourly average is valid if the CEMS has collected at least one data point in each quadrant of the clock hour in which the emission unit was operated. Baghouse inlet temperature data, which demonstrates compliance with dioxin/furan standards, includes one-minute averages comprised of four 15-second readings and three-hour rolling averages calculated as the average of 180 successive one-minute average temperatures. Periods of time when one-minute averages are not available should be ignored when calculating three-hour rolling averages. When one-minute averages become available, it should be added to the previous 179 minutes. Also included in the collection data are digital alarms for temperature exceedances.

7.3 Data Reporting

Cemex is required to submit an Excess Emissions and Continuous Monitoring System Performance Report and all summary reports. These reports shall be delivered or postmarked by the thirtieth (30th) day of the end of the reporting period. Cemex will submit a summary

report semiannually via the Compliance and Emissions Data Reporting Interface by the sixtieth (60th) day of the following month for the reporting period.

The performance report shall contain the following information:

- The date and time identifying each period during which the CMS was inoperative.
- The date and time identifying each period during which the CMS was out-of- control.
- The specific identification of each period of excess emissions and parameter monitoring exceedances.
- The specific identification of each time period of excess emissions and parameter monitoring exceedances.
- Nature and cause of any malfunction.
- The corrective action taken, or preventative measures adopted.
- The nature of the repairs or adjustments to the CMS that was inoperative or out- of-control.
- The total process operating time during the reporting period.

If the total excess emissions or process/control system parameter exceedances periods are less than 1% of the total operating time for the reporting period and the CMS total downtime is less than 5% of the total operating time for the reporting period, then only the summary report shall be submitted, unless required by the Administrator.

If the total excess emissions or process/control system parameter exceedances periods are greater than 1% of the total operating time for the reporting period or the CMS total downtime is greater than 5% of the total operating time for the reporting period, then both the excess emissions and CMS performance report and the summary report shall be submitted.

The summary report shall be entitled “Summary Report – Gaseous and Opacity Excess Emission and Continuous Monitoring System Performance” and contain the following information:

- The company name and address.
- An identification of each hazardous air pollutant (HAP) monitored.
- The beginning and ending dates of the reporting period.
- A brief description of the process units.
- The emission and operating parameter limitations.
- The monitoring equipment manufacturer(s) and model number(s).
- The date of the latest CMS certification or audit.
- The total operating time during the reporting period.
- An emission data summary, including;
  - Excess emissions, reported in minutes for opacity and hours for gases.
  - Total excess emission period expressed in percentage of the total operating time for the reporting period.
  - Breakdown of total excess emission for the reporting period into those due to startup/shutdown, control equipment problems, process problems, other known causes, and other unknown causes.

- CMS performance summary, including;
  - Total CMS downtime, reported in minutes for opacity and hours for gases.
  - Total CMS downtime period expressed in percentage of the total operating time for the reporting period.
  - Breakdown of total CMS downtime for the reporting period into those due to monitoring equipment malfunctions, non-monitoring equipment malfunctions, quality assurance/quality control calibrations, other known causes, and other unknown causes.
- A description of any changes in CMS, processes, or controls since the last reporting period
- The name, title, and signature of the responsible official who is certifying the accuracy of the report; and
- The date of the report.

7.4 Data Backup

The DAHS software polls the PLC to obtain the averages and records them in the DAHS computer. The DAHS computer data files should be backed-up periodically onto a CD or equivalent storage device.

Records shall be retained for at least five years following the date of each occurrence, measurement, maintenance, corrective action, report, or record. At a minimum, the most recent two years of data shall be retained on site. The remaining three years of data may be retained off site. Such files may be maintained on a permanent storage device, such as a CD.

Appendix A Maintenance Checklists

Figure A-1: CEMS Preventative Maintenance Checklist

CEMEX		**CEM SYSTEM**				
Preventative Maintenance Check List						
*****Record Data, Values or Conditions*****						
Technician:		Week of: _____				
		Mon.	**Tues.**	**Wed.**	**Thurs.**	**Fri.**
Acceptable Value						
	Cooler Opacity Monitor					
	Check Current Opacity Reading					
	Check for Alarms					
	Cooler SICK PM Monitor					
	Current PM Reading					
<30%	Current Contamination Value					
	Check for Alarms					
<25	Laser Amps					
	KILN SICK PM Monitor					
	Current PM Reading					
<30%	Current Contamination Value					
<25	Laser amps					
	Check for Alarms					
	Air Cleanup Panel					
>110 psi	Check Plant Supply Pressure (PG1)					
>100 psi	Check incoming air pressure (FR2)					
	Inspect Filters for Moisture/Oil (CF1)					
	Inspect Filters for Moisture/Oil (CF2)					
	Inspect Filters for Moisture/Oil (FR2)					
	Inspect Filters for Moisture/Oil (F2)					
	N2 Generation Panel					
>95 psi	Check Plant Supply Pressure (PR1)					
>95 psi	Check Plant Supply Pressure (PR2)					
>95 psi	Check Plant Supply Pressure (FR1)					
	Inspect Filters for Moisture/Oil (FR1)					
	Inspect Filters for Moisture/Oil (F1)					
	CEM GAS BOTTLES					
>500 psi	Record Zero Cal Cyl. Pressure					

>500 psi	Record O2/THC Span Cyl. Pressure					
>500 psi	Record Hydrogen Cyl Pressure					
	SICK THC KILN					
60 psi +/- 8 psi	Check Instr. Regulator Press					
40 psi +/- 2 psi	Check Zero Air Regulator Press					
75-90 LPH	Current THC Concentration					
0-50 ppm	Check for Alarms					
	MCS100E					
PPM	Current NO Concentration					
PPM	Current SO2 Concentration					
PPM	Current CO Concentration					
%	Current CO2 Concentration					
%	Current O2 Concentration					
100%-60%	Energy %					
120 - 700 LPH	Sample Flow					
200 °C +/- 10 °C	Heated Umbilical Line Temp (E41.1)					
200 °C +/- 10 °C	Heated Probe Temp (E42.1)					
200 °C +/- 10 °C	Heated Probe Tube Temp (E42.2PR)					
200 °C +/- 10 °C	Heated Pump Temp (M1)					
	Check Fans and Filters					
	Check for Alarms					
	Apex Hg CEMS					
	Check Condensate Bottles/Check for alarms					
100-500 sccm	Sample Flow A/B					
<15" Hg	Sample Vacuum A/B					
	Verify Toughbook/APEX is synchronized with DAHS time					
	Trap Changeout Required					
	Run data uploaded to shared drive					
	All Data Results up to date					
	Kiln Opacity monitor					
	Check Current Opacity Reading					

	Check for Alarms					
	Kiln Stack Flow Monitor					
	Current Flow Reading					
	Check for Alarms					
Spotlight	**Daily Calibrations**					
± 5%	Record Flow Calibration % Error					
± 5%	Record PM Calibration % Error					
± 10%	Record CO Calibration % Error					
± 1%	Record CO2 Calibration % Error					
± 5%	Record NO Calibration % Error					
± 1%	Record O2 Calibration % Error					
± 5%	Record THC KILN CAL % ERROR					
± 5%	Record SO2 Calibration % Error					
-0.025	Record NH3 Calibration Error					
± 5%	Record Opacity Calibration % Error					
± 5%	Record Hg Calibration % Error					
	Record Dryer THC CAL Low Error					
	Record Dryer THC CAL High% Error					
± 5%	Record Cooler PM Calibration % Error					
± 5%	Record Cooler Opacity Calibration % Error					
	DRYER GAS BOTTLES					
>200 psi	Record Zero Cal Cyl. Pressure					
>200 psi	Record O2/THC Span Cyl. Pressure Low					
>200 psi	Record O2/THC Span Cyl. Pressure High					
>200 psi	Record Hydrogen Cyl 1 PRES					
	DYER SICK THC GMS 840					
>80 psi	Check Main Air Pressure					
< 0 °C PressDewPnt	Check Air Moisture Level					
80 psi ± 8 psi	Check Instr. Regulator Press					
17.5 psi ± 2 psi	Check Zero Air Regulator Press					
40 l/h	Check Zero Air Flow					
20 psi	Purge Air Pressure					
0-50 ppm	Current THC Concentration					
	TEMP. PROBE H (SL1)					

	TEMP. PROBE T (Straw)					
	TEMP. SAMPLE LINE (HSL2)					
	TEMP. HEATED PUMP					
	Check for Alarms					
	DAILY CEMS LOG					
	Daily CEMS Items Recorded Y/N					
	Daily CEMS log uploaded to Shared drive (monthly) Y/N					
	CEMS COMPRESSOR INSPECTION					
	Visual Compressor Inspection					
	Service Required					

Appendix B Thermocouple Audit Calibration Form

Figure B-1: Thermocouple Calibration Audit Form

Lyons Cement Plant
Baghouse Inlet Temperature Thermocouple
Quarterly Calibration & Inspection

Date: _____ Start Time: _____

Finish Time: _____

Calibrator Serial #: _____ Date Calibrator Certified: _____

Circle Baghouse: MAIN or HART Recertification Due By: _____

NOTE: All temperature readings should be accurate to ± 5 °F.

Put in maintenance mode by pushing the MBH D/F Temp button on the PLC Controls screen. Push the button again when calibration is complete.

1. Remove Thermocouple: Inspect and Clean
2. Reinstall Thermocouple
3. Remove TC wires from Thermocouple. Induce Temperatures using Calibrator.

   100° Call Control Room, They read _____

   300° Call Control Room, They read _____

   595° Call control Room, They read _____

4. Put everything back in service. Call control room and see what they are reading _____
5. Check with your temperature indicator. Your reading is _____

Comments: _____

_____

Sign: _____

Appendix C Spare Parts List

Spare parts will be kept at the discretion of the CEMS technician and the Electrical Department. Recommended parts lists can be located in the CEMs shelters within the equipment operator's manuals.

Continuous Monitoring System
Quality Assurance Program
Revision Log

Revision No.	Revision Date	Revised Sections	Notes
Original	June 2015	NA	Original based on CEMS As Built drawing set.
Revision 1	August 2015	All	Incorporating comments and general updating
Revision 2	December 2015	Various	Incorporating comments and general updating
Revision 3	January 2017	Chapter 6	Add RATA guidelines for THC, NH3, Flow, Hg, Calculations
Revision 4	May 2019	Chapter 2, 5-7	Replace Dryer THC Analyzer - ABB AO204 substituted in favor of SICK FIDOR GMS-840
Revision 5	May 2024	All	Updated entire document

END OF DOCUMENT